

Errata

Corrections to docs/paper.md ("Scaling Laws Do Not Predict Adversarial Robustness in LLM-Based Security Detection Systems"). This is a separate document — the paper itself is not edited. All three items below will be folded into the v2 revision.

Dated 2026-09-05.

1. Meta-Resilience was never computed for Table 1

Section 3.2 declares DABS as five weighted components, including:

Meta-Resilience | 10% | Resistance to adversarial reasoning injection via log fields

Table 1 (§4.1), however, only reports four components — Coverage, Resilience, Hardening, Consistency — with no Meta-Resilience column. The codebase confirms why: no call site that produced the five-model scaling result ever populated `meta_resilience` in a technique result, so the component was always absent for that experiment.

The actual DABS scores in Table 1 were therefore computed on the renormalized 4-component form, with effective weights:

Component	Nominal (paper)	Effective (Table 1, meta absent)
Coverage	30%	33.3%
Resilience	25%	27.8%
Hardening	20%	22.2%
Consistency	15%	16.7%
Meta-Resilience	10%	— (never populated)

This renormalization was never disclosed in the paper. The codebase now names this exact weighting `dabs_v1` (`engine/dabs_scorer.py:WEIGHT_PROFILES`) so it can be reproduced precisely, and every DABS result now records `weight_profile`, `weights_nominal`, and `weights_effective` explicitly instead of silently renormalizing.

Conclusion unaffected. Table 1's numbers do not change — this corrects what the paper says about how they were computed, not the numbers themselves.

2. Two model parameter counts were wrong

Table 1 (§3.3) listed qwen2.5:7b at 7.0B and qwen2.5:14b at 14.0B — rounded, unsourced figures. Verified against the official Hugging Face model cards:

Model	Paper value	Corrected value	Source
qwen2.5:7b	7.0B	7.61B	https://huggingface.co/Qwen/Qwen2.5-7B
qwen2.5:14b	14.0B	14.7B	https://huggingface.co/Qwen/Qwen2.5-14B

The other three model's figures (phi3.5:latest 3.8B, mistral:7b 7.0B, llama3.1:8b 8.0B) were re-verified against their own official cards and are unchanged.

Refitting the same power law ($DABS = a \times P^b$, log-linear regression, same 5 DABS scores from Table 1) with the corrected parameter counts:

Fit	Equation	R ²
Published	$DABS = 65.36 \times P^{-0.087}$	0.0557
Corrected	$DABS = 64.67 \times P^{-0.080}$	0.0530

Conclusion unaffected. The corrected R² (0.0530) is within rounding distance of the published value (0.0557) — parameter count explains essentially none of the variance either way.

3. The weekly automated battle badge never reflected a real result

`.github/workflows/weekly-duel.yml` ran every Monday from 2026-04-25 through 2026-08-31 (21 runs) using Groq model IDs (`llama-3.1-70b-versatile` as Attacker, `mixtral-8x7b-32768` as Defender) that were decommissioned on Groq's side. Every Defender call failed; every technique recorded zero rounds; the workflow's own guard only checked "did any technique avoid raising an exception," which a zero-round battle satisfies, so it committed a 0.0% / 0-0 badge to `README.md` every single week without ever raising an error.

This automation is entirely separate from the scaling-law experiment — its output was never read by `docs/paper.md` — so it does not affect any number in the paper. It is listed here because it was discovered during this audit and is a factual correction about a claim the

README made ("Last Weekly Battle") that was never true.

Fixed: the badge is paused (README.md), scripts/weekly_battle.py now refuses to run against a known-decommissioned model ID and fails the job (instead of committing) on any technique error or an all-zero result, and raw weekly output is now saved to output/benchmarks/weekly/ (versioned in git) instead of only a gitignored path.

4. The original runs used live threat-intelligence enrichment, not seed-controlled

Live threat-intelligence enrichment (URLhaus/Feodo, added 2026-04-25) was active during the original 5-model runs and is not seed-controlled; the paper's reproducibility claim (seed=42) therefore did not hold. Benchmarks now run with `--threat-intel off` by default; the mode is recorded in every output JSON.

5. The paper's seed=42 claim was false — seeding did not exist yet when Table 1 was produced

The paper states (§3.3) that all runs used a fixed seed of 42. Seeding was added to the agents on 2026-05-10 (commit e30fdb0), two days after the paper draft (eada793, 2026-05-08) and after the Zenodo DOI was minted (d6af684, 2026-05-09). Verified directly: at the commit that introduced the scaling-law feature and produced Table 1 (226aed4, 2026-05-08), neither AttackerAgent nor DefenderAgent had a seed parameter, and no `ollama.chat()` call passed a "seed" key in its options. The Table 1 results were generated without any seed; the reproducibility claim was false for the published data.

This is the leading candidate explanation for why this audit's reproduction (same 5 models, `--seed 42, --threat-intel off`) does not land on the original Table 1 values: the original run was never deterministic in the first place. A dedicated isolation test (mistral:7b, unseeded, 3 independent runs, see docs/scaling_v2_results.md) is queued to check whether removing the seed alone accounts for the gap.

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